

SUBJECT NAME: COMPUTER NETWORKS**Code: 23ACS16****Question Bank****UNIT-I**

PART-A		2 MARKS	CO	BTL
1.	What is a computer network?		CO1	BTL1
2.	Define PAN, LAN, MAN, and WAN.		CO1	BTL1
3.	What is an internetwork?		CO1	BTL1
4.	What is a network protocol?		CO1	BTL1
5.	What are the design goals of a computer network?		CO1	BTL1
6.	What is protocol layering?		CO1	BTL1
7.	What is reliability in computer networks?		CO1	BTL1
8.	What are service primitives?		CO1	BTL1
9.	List the layers of the OSI reference model.		CO1	BTL1
10.	Differentiate between the OSI model and the TCP/IP model.		CO1	BTL4
PART-B		10 MARKS		
1.	Explain the different types of computer networks with suitable examples.		CO1	BTL2
2.	Describe the design goals of computer networks in detail.		CO1	BTL2
3.	Explain protocol layering with advantages and disadvantages.		CO1	BTL2
4.	Explain connection-oriented and connectionless services.		CO1	BTL2
5.	Explain service primitives with examples.		CO1	BTL2
6.	Describe the relationship between services and protocols.		CO1	BTL2
7.	Explain the OSI reference model with neat diagram.		CO1	BTL2
8.	Explain the TCP/IP reference model with neat diagram.		CO1	BTL2
9.	Compare the OSI and TCP/IP reference models.		CO1	BTL4
10.	Explain the criticism (critique) of the OSI model and TCP/IP model.		CO1	BTL5
UNIT-II				
PART-A		2 MARKS	CO	BTL
1.	What is the Data Link Layer?		CO2	BTL1
2.	What is guided transmission media?		CO2	BTL1

3.	Define twisted pair cable.	CO2	BTL1
4.	What is optical fiber?	CO2	BTL1
5.	What is framing?	CO2	BTL1
6.	What is flow control?	CO2	BTL1
7.	What is error detection?	CO2	BTL1
8.	What is CRC?	CO2	BTL1
9.	What is Ethernet?	CO2	BTL1
10.	What is CSMA/CD?	CO2	BTL1
<u>PART-B</u> 10 MARKS			
1.	Explain different types of guided transmission media.	CO2	BTL2
2.	Explain the functions of the Data Link Layer.	CO2	BTL2
3.	Explain framing methods with examples.	CO2	BTL2
4.	Explain error detection and error correction techniques.	CO2	BTL2
5.	Explain flow control using Stop-and-Wait and Sliding Window protocols.	CO2	BTL2
6.	Explain channel allocation methods.	CO2	BTL2
7.	Describe multiple access protocols in detail.	CO2	BTL2
8.	Explain collision-free protocols.	CO2	BTL2
9.	Explain Ethernet architecture and MAC sublayer.	CO2	BTL2
10.	Explain Fast Ethernet, Gigabit Ethernet, 10-Gigabit Ethernet, 40-Gigabit Ethernet, and 100-Gigabit Ethernet.	CO2	BTL2
UNIT-III			
	PART-A 2 MARKS	CO	BTL
1.	What is packet switching?	CO3	BTL1
2.	What is a virtual circuit?	CO3	BTL1
3.	What is a datagram network?	CO3	BTL1
4.	Define routing.	CO3	BTL1
5.	What is flooding?	CO3	BTL1
6.	What is congestion?	CO3	BTL1
7.	What is packet fragmentation?	CO3	BTL1

8.	What is IPv4?	CO3	BTL1
9.	What is IPv6?	CO3	BTL1
10.	What is ICMP?	CO3	BTL1
<u>PART-B</u>		10 MARKS	
1.	Explain the design issues of the Network Layer.	CO3	BTL2
2.	Explain store-and-forward packet switching.	CO3	BTL2
3.	Compare virtual circuit and datagram networks.	CO3	BTL4
4.	Explain shortest path routing algorithm.	CO3	BTL2
5.	Explain Distance Vector Routing algorithm.	CO3	BTL2
6.	Explain Link State Routing algorithm.	CO3	BTL2
7.	Explain congestion control techniques.	CO3	BTL2
8.	Explain IPv4 addressing and packet format.	CO3	BTL2
9.	Explain IPv6 features and packet format.	CO3	BTL2
10.	Explain OSPF and BGP routing protocols.	CO3	BTL2
UNIT-IV			
PART-A		2 MARKS	CO BTL
1.	What is the Transport Layer?	CO4	BTL1
2.	Define transport service.	CO4	BTL1
3.	What are Berkeley sockets?	CO4	BTL1
4.	What is socket programming?	CO4	BTL1
5.	What is multiplexing?	CO4	BTL1
6.	What is congestion control?	CO4	BTL1
7.	What is UDP?	CO4	BTL1
8.	What is TCP?	CO4	BTL1
9.	What is a TCP segment?	CO4	BTL1
10.	What is a three-way handshake?	CO4	BTL1
<u>PART-B</u>		10 MARKS	
1.	Explain the services provided by the Transport Layer.	CO4	BTL2
2.	Explain transport service primitives.	CO4	BTL2

3.	Explain Berkeley sockets with examples.	CO4	BTL2
4.	Explain Internet transport protocols (TCP and UDP).	CO4	BTL2
5.	Compare TCP and UDP.	CO4	BTL4
6.	Explain the TCP service model.	CO4	BTL2
7.	Explain the TCP segment header with neat diagram.	CO4	BTL2
8.	Explain TCP connection establishment using three-way handshake.	CO4	BTL2
9.	Explain TCP connection release.	CO4	BTL2
10.	Explain error control, flow control, and congestion control in the Transport Layer.	CO4	BTL2
UNIT-V			
	PART-A	2 MARKS	CO
1.	What is the Application Layer?	CO5	BTL1
2.	What is electronic mail?	CO5	BTL1
3.	What is SMTP?	CO5	BTL1
4.	What is HTTP?	CO5	BTL1
5.	What is HTTPS?	CO5	BTL1
6.	What is a web browser?	CO5	BTL1
7.	What is a web server?	CO5	BTL1
8.	What is a proxy server?	CO5	BTL1
9.	What is a Content Delivery Network (CDN)?	CO5	BTL1
10.	What is a Peer-to-Peer (P2P) network?	CO5	BTL1
	<u>PART-B</u>	10 MARKS	
1.	Explain the architecture and services of electronic mail.	CO5	BTL2
2.	Explain the message transfer process in e-mail.	CO5	BTL2
3.	Explain the architecture of the World Wide Web.	CO5	BTL2
4.	Explain HTTP and HTTPS in detail.	CO5	BTL2
5.	Explain static web pages and dynamic web pages.	CO5	BTL2
6.	Explain web privacy and Internet traffic.	CO5	BTL2
7.	Explain web proxies and server farms.	CO5	BTL2

8.	Explain Content Delivery Networks (CDNs).	CO5	BTL2
9.	Explain Peer-to-Peer (P2P) networks with advantages and applications.	CO5	BTL2
10.	Explain the evolution of the Internet.	CO5	BTL2